



CORDIS Results Pack on Horizon in Bulgaria

A thematic collection of innovative EU-funded research results

April 2025

Supporting pioneering research, from drone airlines to cancer care



Research and
Innovation

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Supporting pioneering research, from drone airlines to cancer care

For over 40 years, the European Union has supported cutting-edge science, research and innovation. By driving scientific progress, the EU R&I programme helps to tackle some of the most important challenges facing Europe, from cancer to climate change.

This CORDIS Results Pack on Horizon in Bulgaria highlights the impact of EU funding in fostering scientific innovation, increased competitiveness, employment opportunities, and the growth and scale-up of innovative businesses.

Bulgaria has a long history of scientific innovation. Artefacts recovered from the Varna Necropolis represent the oldest known gold treasure and jewellery in the world, reflecting an active metallurgy industry that existed in this region nearly 7 000 years ago.

Bulgaria continues to uphold this tradition of science, technology, industry and commerce to this day. Its scientists and innovators have produced major advances in chemistry, physics, pharmacology, microbiology, aeronautics, astronomy, spaceflight, and much more. The first digital electronic computer, jet engines, airbags, the tuberculosis vaccine, and commercial yoghurt production all owe their existence to the contributions of Bulgarian researchers.

Through its flagship funding programmes, most recently Horizon 2020 (2014-2020) and Horizon Europe (2021-2027), the EU has invested over EUR 500 million across almost 700 research and industry organisations in Bulgaria¹. This support comes from various initiatives, including the [European Research Council](#), [Marie Skłodowska-Curie Actions](#), [European Innovation Council](#), and more.

The nine projects in this Pack highlight the breadth of the research supported through Horizon in Bulgaria. They include work improving women's access to medicine via advanced screening technologies, the use of lasers to produce materials with novel surface properties, uncrewed drone airlines for short-haul logistics, improved diagnostics, eco-friendly bioplastics, forensic techniques to uncover clandestine bomb labs, and more.

Together, these projects showcase the continuing importance of Bulgaria's contributions to science, research and innovation, and the country's central role in tackling major challenges of our time.

¹ Data retrieved from Horizon Dashboard March 2025, see bit.ly/HorizonBulgaria

AI and thermal imaging assist rapid disease diagnosis

A clever technique that incorporates thermal imaging and artificial intelligence from the EU-funded AI-CARE project could transform the detection of a difficult-to-diagnose condition that affects millions of people.



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Peripheral arterial disease (PAD) involves the narrowing or blockage of vessels that carry blood from the heart to the legs. Although less well-known than other vascular conditions such as heart disease or stroke, the condition is widespread – and poorly diagnosed.

“More than 230 million people suffer from PAD globally,” says AI-CARE project coordinator Georgi Kadrev, co-founder and CEO of [Kelvin Health](#) in Bulgaria.

“Every year, more than 22 million people develop the most severe form of the disease, called critical limb ischemia, which markedly reduces blood flow. This can lead to amputations, and has a fatality rate of 70 % within five years of diagnosis.”

Clinical need for innovations

Current diagnosis methods include taking blood pressure measurements at various points in the ankle, which can be

imprecise. In particular, diagnosing patients with diabetes can be challenging due to arterial calcification and a lack of sensitivity in the legs. More invasive methods, including angiography (X-rays used to check blood vessels), are only used in cases suspected to be serious.



Any abnormalities will be picked up by the algorithms, telling us of the presence of any arterial blood supply problems.

For Kadrev and his colleagues, there was a clear clinical need for a simple, accurate and affordable diagnosis of this silent disease. “Our concept was to develop a system that applies a portable thermal imaging camera that captures body thermodynamics,” explains Kadrev. “This generates a series of images, which are then segmented and analysed using artificial intelligence (AI) image recognition algorithms.”

Kadrev demonstrates the technology by attaching the thermal camera to his mobile phone and taking images of himself. Any anomalies related to the peripheral vascular system would be detected using the system’s machine learning algorithms.

“The whole premise of the concept is that our local body temperature is related to blood flow,” explains Kadrev. “Any abnormalities will be picked up by the algorithms, telling us of the presence of any arterial blood supply problems.”

Bringing medical innovation to market

Bringing the technology to market requires traversing what is known as the ‘valley of death’, with significant investment needed to translate an idea with promise into a viable marketable product.

“We need to externally validate our prototype in a clinical setting,” says Kadrev. “Just getting to this stage requires a great deal of administrative work, intellectual property analysis, and awareness of issues such as medical data protection. We also need to develop a market access planning strategy.”

The AI-CARE project provided Kelvin Health with the support it needed to reach this critical milestone. “We are now preparing

for actual clinical validation and starting to collect clinical data,” adds Kadrev. “All this is a prerequisite to commercialisation.”

Regular screening of at-risk patients

The project has also helped to raise the profile of innovation in Bulgaria. “We have been able to work with key opinion leaders,” notes Kadrev. “We were recently invited to the [American Heart Association Scientific Sessions](#) in Chicago last November, and the [LINC conference](#) in Leipzig, the leading European conference on arterial interventions.”

Horizon’s support for European innovation ecosystems is also key to fostering a start-up culture. This is especially true where existing venture capital is not oriented towards research solutions, especially in the healthcare space, because of the time and risks involved. “This has helped to put us on the map,” remarks Kadrev.

The ultimate goal is to bring this innovation into clinical settings, and to establish the technique as an effective and cost-effective means of screening for PAD. “We could see a situation where patients at risk of [PAD](#) – heavy smokers, people with diabetes – are screened twice a year, helping to save resources and save lives,” adds Kadrev.

PROJECT

AI-CARE – Thermography AI – cornerstone in future preventive healthCARE

COORDINATED BY

Kelvin Health in Bulgaria

FUNDED UNDER

Horizon Europe – European innovation ecosystems

CORDIS FACTSHEET

cordis.europa.eu/project/id/101113791

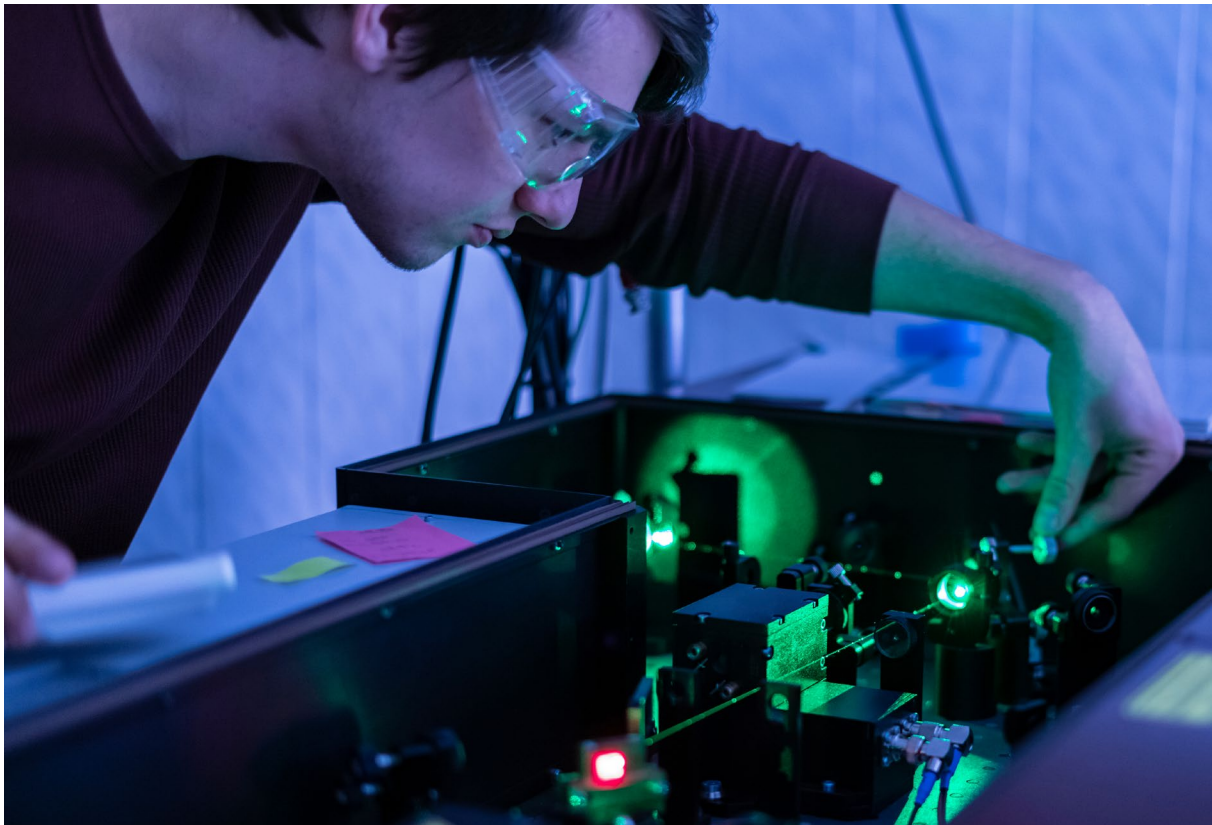
PROJECT WEBSITE

kelvin.health



Scientific exchanges strengthen nanotech laser modelling

More accurate modelling of laser-matter interaction led by the EU-funded ATLANTIC project could lead to new industrial applications and help establish Bulgaria as an important research hub in the field.



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In nature, nanoscale structures on the surface of materials give rise to properties such as the vibrant colours of butterfly wings and the hydrophobic qualities of a lotus leaf. In the lab, scientists can replicate this process using bursts of intense laser light, offering novel materials with desirable properties.

"Short pulse laser technology could provide solutions for many industrial applications," explains ATLANTIC project coordinator

[Tzvetta Apostolova](#) from the [New Bulgarian University](#). "These include microprocessing of biopolymers and metal alloys for medical stent fabrication, removing thin layers from solar cells, drilling micro vias on printed circuit boards, as well as forming nanostructures used for memory storage in electronic components, bone implants."